Draft cover letter: for author review; not sent

Dear PLOS ONE Editors,

Please consider the Research Article “Longitudinal motor-imagery EEG decoding: Frozen baselines, supervised recalibration, and task-measurement reliability.”

This secondary analysis compares decoding updates in 18 participant records across six sessions of Kumar2024. The acquisition study investigated online covariance recentering and Riemannian class-prototype updating, with a separate pseudo-online CSP extension. Our analysis distinguishes classifier-only from full CSP/LDA refitting under the same 19/20-label current-session support budget and examines whether available task measurements justify individual update decisions.

The predefined primary comparison in 12 participants favored full refitting over frozen decoding by 7.42 percentage points (95% participant-bootstrap interval 2.49–12.48). The secondary same-label-budget comparison favored full refitting over LDA-only updating by 5.47 points (2.17–8.91). All neural seeds, negative outcomes and class-collapsed predictions are retained. Weak EEGNet source-query performance limits interpretation of its later target result. QC sensitivity, incomplete class coverage, cue-window phase uncertainty and prior test exposure are reported explicitly. The additional paired tests were introduced after review and are labeled exploratory, with multiplicity shown.

The supporting files provide scientific methods, saved-result inputs/code, figure data and a separate provenance appendix. Original code is MIT licensed; original manuscript and derived content are CC BY 4.0 within the stated scope. The applicable ethical basis for this secondary analysis and final personal author approval remain to be completed before this letter is used.

Sincerely,

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Author-only note: complete the ethics basis, personal final review and actual portal-specific entries. This letter has not been sent.